Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Dr. Snehal Andhare

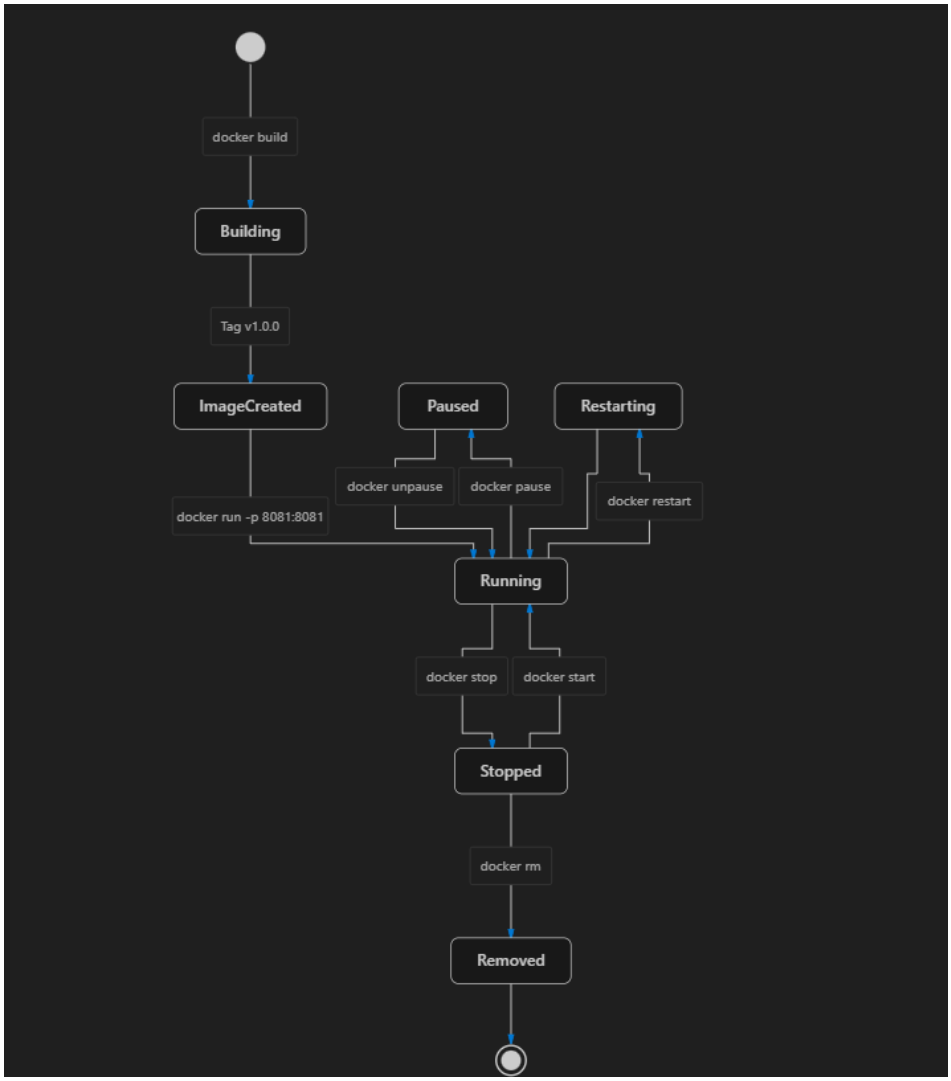
Student Name	Vicky Pukale
Roll Number	23102A0033
Grade and Subject Teacher's Signature	

Task Number	11
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	Docker Image and Container Lifecycle: Create a Dockerfile for the Loan Application Tracking Portal. Build and tag the image, run the container, map ports, inspect logs, stop/restart/remove containers and document the complete container lifecycle.

Deliverable:

Container Lifecycle & Execution Guide:

1. Container Lifecycle State Diagram



2. Step-by-Step Docker Command Log

```
D:\Vicky_Project\DevOps\DevOps_Project>docker build -t loan-portal:v1.0.0 -t loan-portal:latest .
[+] Building 96.2s (19/19) FINISHED                                                                 docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 1.24kB                                                                              0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jre-alpine                                    3.2s
 => [internal] load metadata for docker.io/library/maven:3.9.6-eclipse-temurin-17-alpine                            3.2s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                              0.0s
 => [auth] library/maven:pull token for registry-1.docker.io                                                        0.0s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 144B                                                                                   0.0s
 => [builder 1/6] FROM docker.io/library/maven:3.9.6-eclipse-temurin-17-alpine@sha256:ffddac7b041013580 48f872fba  14.1s
 => => resolve docker.io/library/maven:3.9.6-eclipse-temurin-17-alpine@sha256:ffddac7b041013580 48f872fba9d978b2a1  0.0s
 => => sha256:538fb58edc7f673e614b2e6cc20861b1677d505e6387bbb9568b8680db36f5f1 166B / 166B                          0.3s
 => => sha256:d77f74971d2cb36864147a4a52c5557e0ff834f8c2fe569490d431ead453cc0c 356B / 356B                          0.3s
 => => sha256:0eee4ba058f437133e846346ab57c2a8024666cf58bbf28b3dc7aa600bbc9853 860B / 860B                          0.5s
 => => sha256:2575e8f1a97f8a73f2bf2b31eba5ac4a21b62e56c09e9351c79fdc029e58b334 9.48MB / 9.48MB                      1.8s
 => => sha256:9312175dbfcff354e2b62118a8893aa8ea1e5e2c18740e3e4197a691bef75ad0 2.63MB / 2.63MB                      1.9s
 => => sha256:f852f18c8554898c2508d8273094bbe0d0a6d5fc3175658c756845270a1f5cca 717B / 717B                          0.3s
 => => sha256:71edf4fdf61d7c4bcebfbe05d7693194 7068e88ea252a9ccab7c03b3b131f074 188B / 188B                         0.3s
 => => sha256:a08b7b3a74c4535b6568ccd03b6b5fb1de2dbd6cb55f506618b3471acf06b528 144.33MB / 144.33MB                  9.9s
 => => sha256:a3fd38fd7cf5b8d60c92e1aa46a24527229fb51b451491d35a7028a8a1d0aba4 13.14MB / 13.14MB                     2.8s
 => => sha256:4abcf20661432fb2d719aaf90656f55c287f8ca915dc1c92ec14ff61e67fbaf8 3.41MB / 3.41MB                      0.8s
 => => extracting sha256:4abcf20661432fb2d719aaf90656f55c287f8ca915dc1c92ec14ff61e67fbaf8                            0.2s
 => => extracting sha256:a3fd38fd7cf5b8d60c92e1aa46a24527229fb51b451491d35a7028a8a1d0aba4                            0.5s
 => => extracting sha256:a08b7b3a74c4535b6568ccd03b6b5fb1de2dbd6cb55f506618b3471acf06b528                            0.9s
 => => extracting sha256:71edf4fdf61d7c4bcebfbe05d7693194 7068e88ea252a9ccab7c03b3b131f074                           0.0s
 => => extracting sha256:f852f18c8554898c2508d8273094bbe0d0a6d5fc3175658c756845270a1f5cca                            0.0s
 => => extracting sha256:9312175dbfcff354e2b62118a8893aa8ea1e5e2c18740e3e4197a691bef75ad0                            0.1s
 => => extracting sha256:2575e8f1a97f8a73f2bf2b31eba5ac4a21b62e56c09e9351c79fdc029e58b334                            0.1s
 => => extracting sha256:0eee4ba058f437133e846346ab57c2a8024666cf58bbf28b3dc7aa600bbc9853                            0.0s
 => => extracting sha256:d77f74971d2cb36864147a4a52c5557e0ff834f8c2fe569490d431ead453cc0c                            0.0s
 => => extracting sha256:538fb58edc7f673e614b2e6cc20861b1677d505e6387bbb9568b8680db36f5f1                            0.0s
 => [runtime 1/5] FROM docker.io/library/eclipse-temurin:17-jre-alpine@sha256:3c472129dc75a8d1d7a3f2df5b2093a8077  5.2s
 => => resolve docker.io/library/eclipse-temurin:17-jre-alpine@sha256:3c472129dc75a8d1d7a3f2df5b2093a8077e4493def  0.0s
 => => sha256:2b4bfbd732889dc44f8440816407969598d889b521a06a509bfa6f3149e8e0e 2.46kB / 2.46kB                       0.3s
 => => sha256:b13c582bc86c00d52360e98f37b00bc7f02ea3262b11f2e3ded435c6cb82b017 127B / 127B                          0.5s
 => => sha256:647d5c102196b54d2dfccaa7f779a204eb9a388ded7e4053bcce16c387207fcf 47.19MB / 47.19MB                    3.1s
 => => sha256:f67820664a14b831543f8a1c48f926e7d40bdb0f1f4e7a133d53e53559ccbd03 19.44MB / 19.44MB                    3.7s
 => => extracting sha256:f67820664a14b831543f8a1c48f926e7d40bdb0f1f4e7a133d53e53559ccbd03                            0.6s
 => => extracting sha256:647d5c102196b54d2dfccaa7f779a204eb9a388ded7e4053bcce16c387207fcf                            0.7s

D:\Vicky_Project\DevOps\DevOps_Project>docker images | findstr loan-portal
loan-portal:latest                3602aa62e926        458MB          162MB
loan-portal:v1.0.0                3602aa62e926        458MB          162MB
```

2: Run the Container with Port Mapping

```
D:\Vicky_Project\DevOps\DevOps_Project>docker run -d --name loan-portal-app -p 8081:8081 loan-portal:v1.0.0
334ff021008b0b925a3529d62cd5e1a3dda5ff67259a2cc8ae3768a26a301fdf

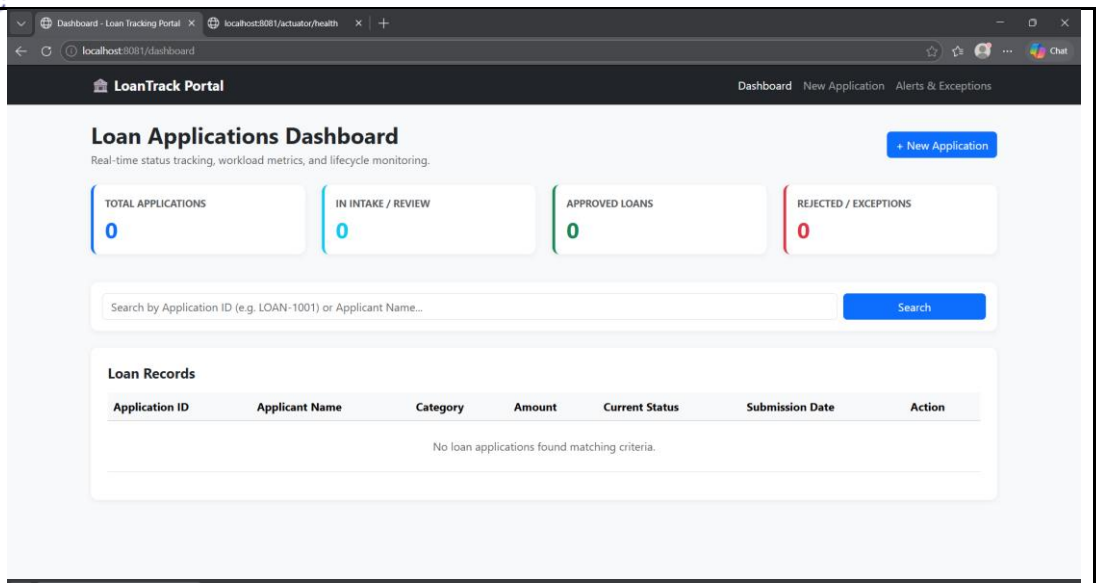
D:\Vicky_Project\DevOps\DevOps_Project>docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS                    PORTS                                       NAMES
334ff021008b   loan-portal:v1.0.0    "java -jar app.jar"      14 seconds ago   Up 13 seconds (healthy)   0.0.0.0:8081->8081/tcp, [::]:8081->8081/tcp   loan-portal-app
```

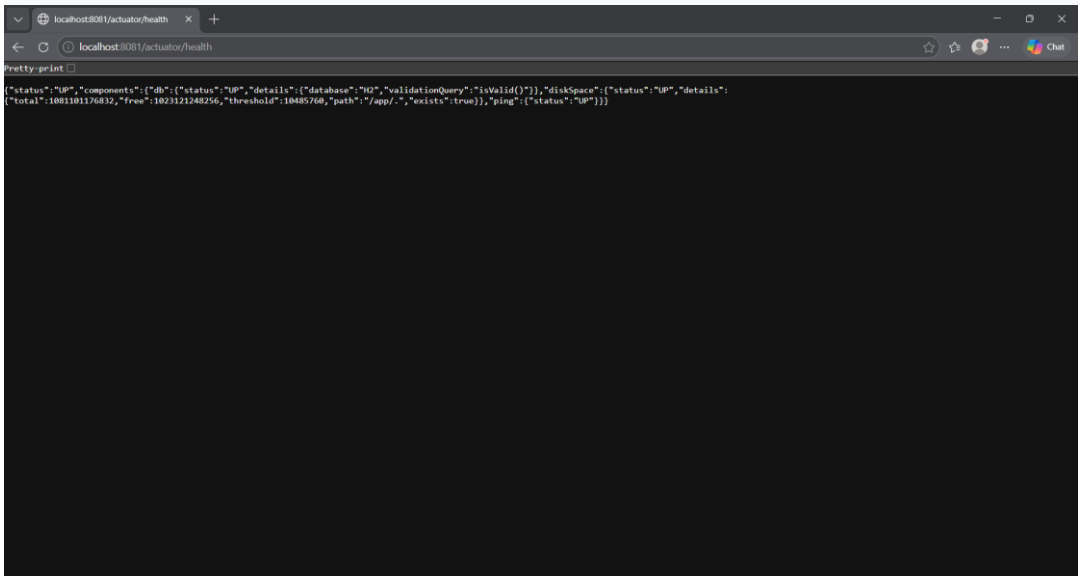
3: Inspect Live Container Logs

```
D:\Vicky_Project\DevOps\DevOps_Project>docker logs --tail 50 loan-portal-app

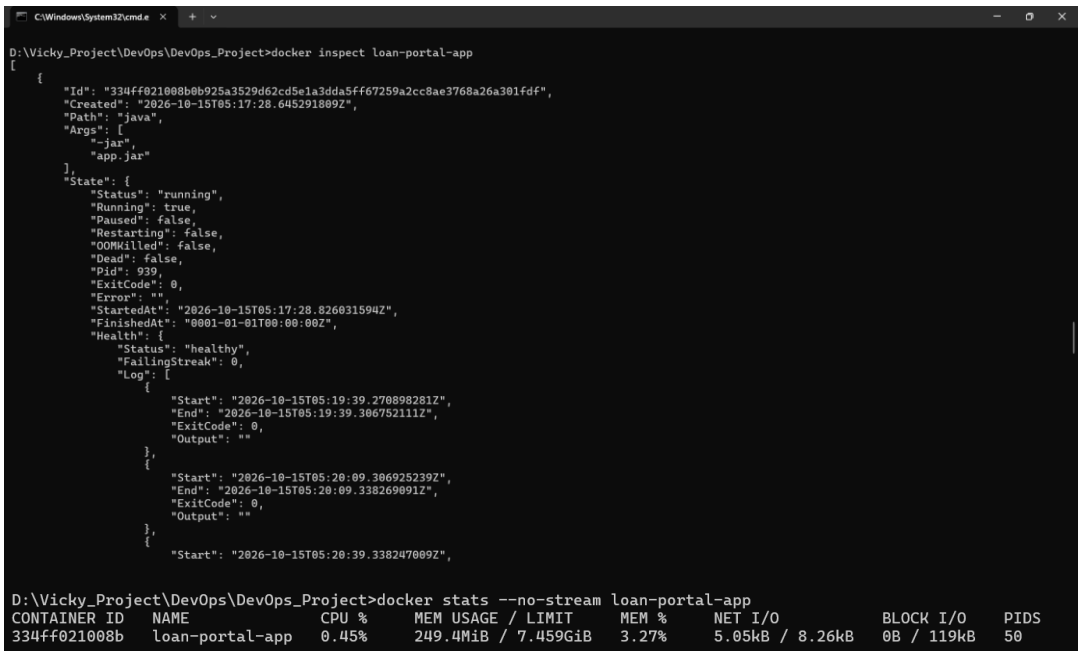
  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v3.2.3)

2026-10-15T05:17:29.880Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] c.loanportal.LoanApplicationPortalApp    : Starting LoanApplicationPortalApp v1.0.0-SNAPSHOT using Java 17.0.20.1 with PID 1 (/app/app.jar started by appuser in /app)
2026-10-15T05:17:29.884Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] c.loanportal.LoanApplicationPortalApp    : No active profile set, falling back to 1 default profile: "default"
2026-10-15T05:17:31.034Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-10-15T05:17:31.093Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 47 ms. Found 2 JPA repository interfaces.
2026-10-15T05:17:31.963Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port 8081 (http)
2026-10-15T05:17:31.977Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-10-15T05:17:31.977Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/10.1.19]
2026-10-15T05:17:32.019Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded WebApplicationContext
2026-10-15T05:17:32.021Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2043 ms
2026-10-15T05:17:32.209Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] com.zaxxer.hikari.HikariDataSource       : HikariPool-1 - Starting...
2026-10-15T05:17:32.476Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] com.zaxxer.hikari.pool.HikariPool        : HikariPool-1 - Added connection conn0: url=jdbc:h2:mem:loandb user=SA
2026-10-15T05:17:32.479Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] com.zaxxer.hikari.HikariDataSource       : HikariPool-1 - Start completed.
2026-10-15T05:17:32.494Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] o.s.b.a.h2.H2ConsoleAutoConfiguration    : H2 console available at '/h2-console'. Database available at 'jdbc:h2:mem:loandb'
2026-10-15T05:17:32.785Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] o.hibernate.jpa.internal.util.LogHelper  : HHH000204: Processing PersistenceUnitInfo [name: default]
2026-10-15T05:17:32.860Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] org.hibernate.Version                    : HHH000412: Hibernate ORM core version 6.4.4.Final
2026-10-15T05:17:32.908Z  INFO 1 --- [LoanApplicationTrackingPortal] [           main] o.h.c.internal.RegionFactoryInitiator    : HHH000026: Second-level cache disabled
```





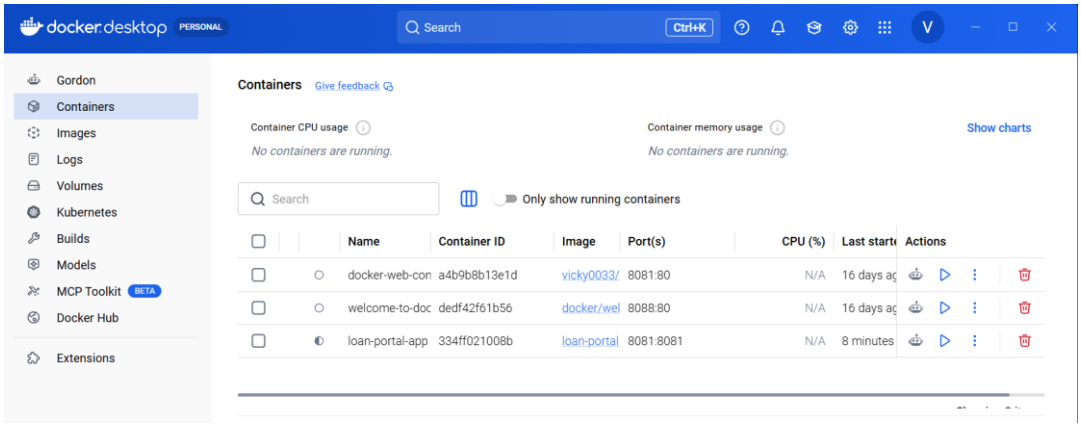
4: Container Inspection & Resource Statistics



5: Container State Transitions (Pause, Unpause, Restart, Stop)

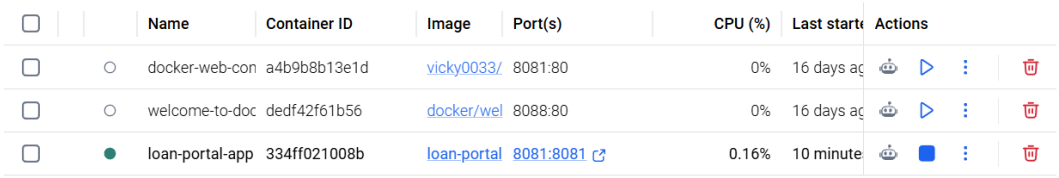
Pause container execution:

```
cmd
docker pause loan-portal-app
```



Unpause container:

```
cmd
docker unpause loan-portal-app
```

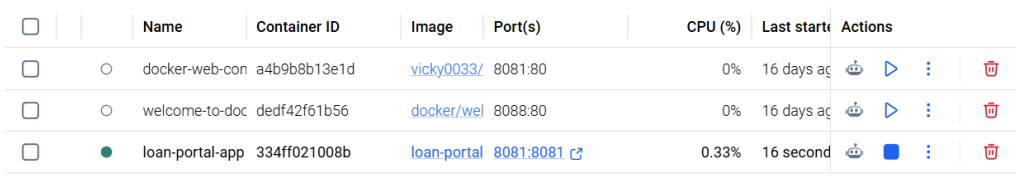



```
D:\Vicky_Project\DevOps\DevOps_Project>docker pause loan-portal-app
loan-portal-app

D:\Vicky_Project\DevOps\DevOps_Project>docker unpause loan-portal-app
loan-portal-app
```

Restart container:

```
cmd
docker restart loan-portal-app
```



- **Stop container:**

```
cmd
```

```
docker stop loan-portal-app
```

```
D:\Vicky_Project\DevOps\DevOps_Project>docker ps -a
CONTAINER ID   IMAGE                      COMMAND                  CREATED          STATUS                      PORTS     NAMES
334ff021008b   loan-portal:v1.0.0         "java -jar app.jar"      12 minutes ago   Exited (143) 25 seconds ago             loan-portal-
app
```

- **Remove container:**

```
cmd
docker rm loan-portal-app
```

```
D:\Vicky_Project\DevOps\DevOps_Project>docker rm loan-portal-app
loan-portal-app
```

3. Commit Docker Configuration to Git

```
D:\Vicky_Project\DevOps\DevOps_Project>git add Dockerfile .dockerignore
warning: in the working copy of '.dockerignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the next time Git touches it

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "ci(docker): add hardened multi-stage Dockerfile with non-root security and healthcheck"
[main 0feb301] ci(docker): add hardened multi-stage Dockerfile with non-root security and healthcheck
 2 files changed, 53 insertions(+)
 create mode 100644 .dockerignore
 create mode 100644 Dockerfile

D:\Vicky_Project\DevOps\DevOps_Project>git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1.07 KiB | 1.07 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/vickypukale/loan-application-tracking-portal.git
   5354812..0feb301  main -> main
```